Somanshu Mohapatra

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SKILLS

Robotics: Forward/Inverse Kinematics (DH Parameters), ROS2, RViz2, MoveIt, Gazebo, Path Planning, Motion Control, PID Control, Sensor Fusion, SLAM.
Electronics: Arduino UNO, Raspberry Pi, STM32, ATmega328P, RP2040, Motor Drivers, Serial Communication (UART, I2C, SPI), LabVIEW, Proteus, NGspice, Pspice.
AI & Vision: AIML, Deep Learning (PyTorch, TensorFlow, CNN, YOLO), OpenCV, OpenVINO, AI Image Generation (Stable Diffusion, AUTOMATIC1111, Illustrious Models).
CAD & Design: SolidWorks, Fusion 360, AutoCAD, Creo Parametric, ANSYS HFSS, Assembly Design, Rapid Prototyping, 3D Printing (FDM/SLA), Bambu Slicer, LightBurn.
Languages: C++, Python, Java, JavaScript, HTML, SQL, Embedded C, MATLAB, GNU Octave.
Soft Skills: Effective Organizer, Team Management, Strong Interpersonal Skills, Problem Solving.

PROJECTS

2.4GHz Microstrip Patch Antenna | *ANSYS HFSS* — Oct' 25

- Designed and simulated a high-frequency microstrip patch antenna optimized for 2.4GHz wireless communication applications.
- Conducted electromagnetic field analysis using ANSYS HFSS to evaluate return loss, gain, and radiation patterns.
- Optimized physical dimensions and substrate selection to achieve superior impedance matching and antenna efficiency.

Dynamic Audio Monitoring and Narration | *Raspberry Pi, Python, AssemblyAI API* — May' 25

- Developed a real-time transcription system for the deaf using Raspberry Pi to provide accessible communication.
- Integrated AssemblyAI API and WebSockets with an I2C SSD1306 OLED display utilizing multi-threading technology.
- Optimized inference speeds on Raspberry Pi 4B by deploying an Intel AI Accelerator, significantly reducing latency.

3-DOF Robotic Arm with Real-Time Control | *MATLAB, SolidWorks, Arduino IDE, Bambu Studio* — Apr' 24

- Constructed a 3-DOF robotic arm utilizing 3D-printed parts to optimize the payload-to-weight ratio for automated tasks.
- Programmed real-time motor control via MATLAB-UART communication, enabling precise joint actuation.
- Targeted future implementation of Forward and Inverse Kinematics to automate complex trajectory planning.

Python-SQL Accounting Software | *Python, MySQL, CLI* — Jun' 23

- Engineered a database-driven accounting CLI application to manage financial records and client transactions.
- Implemented CRUD operations using Python-MySQL connectivity to facilitate seamless data management.
- Provided a robust interface for searching records by client name or date, streamlining data retrieval.

TRAININGS

AutomateX: Mastering PLCs for Industrial Automation | *LPU* — Mar' 25

- Acquired comprehensive knowledge of PLC architecture, ladder logic programming, and industrial automation principles.
- Simulated automated industrial workflows using TwidoSuite to manage sensor inputs and heavy actuator outputs.
- Tested fail-safe mechanisms and sequence patterns to reinforce practical understanding of factory floor operations.

CERTIFICATES

Data Base Management System | *NPTEL* — Aug' 25
Fusioncraft | *Resolute Lab* — Mar' 24

ACHIEVEMENTS

Patent: Method for Enhancing Food Accessibility and Energy Efficiency in Refrigeration — Sep' 25

- Application No.: 202511085342 A

EDUCATION

Lovely Professional University — Phagwara, Punjab
BTech Robotics and Automation; CGPA: 8.38 — *Aug' 23 – Present*

Shri S.N. Siddeshwar Public School — Gurugram, Haryana
Intermediate, PCMB; Percentage: 85.6% — *Mar' 21 – May' 23*

Ryan International School — Gurugram, Haryana
Matriculation; Percentage: 81.6% — *Mar' 20 – May' 21*